

Medium Access Control Protocols

Module 08

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Video: <https://www.youtube.com/watch?v=Yw1HsIBvLLM>

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0:00 science Punjabi University patella today's lecture is on Mac protocols in this lecture we are going

0:07 to discuss in detail about the working and Protocols of medium access sublayer of OSI model now let's

0:16 start the medium access sublayer is the bottom part of data link layer the medium access sub layer is also known as

0:24 Mac that is Media Access Control sublay when a common medium is shared by many stations Mac lab plays a very important

0:33 role without this control several stations transmitting simultaneously could produce gobbled messages the media

0:41 Access Control mechanism standardized by i e are implemented in the max sublayer of the data link layer it provides

0:49 service to the LLC logical link control sublayer and receives service from the physical layer below it the basic

0:58 functions of Max sublayer is the media Access Control error detection and station addressing media Access Control

1:06 procedures are implemented to ensure that every station gets a fair chance to transmit and collisions do not take

1:14 place there are several methods of media Access Control in lands and each method is applicable to specific land topology

1:23 in addition to the basic access procedures the Mac layer also handles the frame D limiting address recognition

1:31 and error checking functions when a number of user stations share a single transmission medium this is called a

1:38 multiple access communication the transmission medium is broadcast in nature so all other

1:45 attached stations to the medium can receive the transmission From Any Given station the physical transmission medium

1:53 of a land is shared by the stations connected on the land let's see how media is

2:01 2 minutes, 1 second shared the two basic approaches to medium Access Control are static and dynamic access control or in other words

2:09 it is called Static Channel allocation and dynamic Channel allocation the dynamic access control is further

2:15 categorized into random access and scheduling the figure here shows various techniques for sharing a transmission

2:24 medium first we'll talk about static Channel allocation in land number of computers are working and only one

2:31 computer is allowed to broadcast on the communication channel the first method of allocating a single Communication

2:38 channel among multiple contending users is by using frequency division multiplexing that is

2:45 fdm if there are n number of users the entire bandwidth is divided into n equal

2:52 partitions and each user is allocated one partition since each user has been allocated a fixed frequency band

3:01 3 minutes, 1 second therefore interference between them is avoided but this scheme is applicable only when number of users are less and

3:11 each has constant traffic when number of users in the land is large with constantly waiting traffic then the

3:19 bandwidth is divided equally into n partitions if the number of users is less than n than a significant portion

3:26 of frequency spectrum is wasted but if the number of users is greater than n then some of them will not get a

3:34 frequency Channel due to shortage of bandwidth in this case channel is set to be inefficient if all the users who want

3:43 to communicate are arranged in one q and number of users are n then the average

3:50 delay T will be n times the meantime delay the static fdm exhibits poor

3:57 performance of Channel next method of Channel allocation is dynamic Channel allocation Dynamic

4:04 Channel allocation method is included in all lands and vans Dynamic Chanel allocation method is able to handle all

4:13 the different types of traffic conditions an order is built up among the Stations of the land so that proper

4:20 chance is given to each station to transmit its data frames but for this some requirements are to be satisfied by

4:28 the land first requirement is that the workstations or terminals are independent and work is generated at a

4:36 constant rate second requirement is that a single channel is available for all communication all stations can transmit

4:45 and receive through it third requirement is that the workstation can begin signal transmission at any time or in a slotted

4:55 time assigned to it fourth requirement is that if two stations are transmitting simultaneously the signals collide with

5:03 each other and resulting signal is gobbled this process is called Collision this Collision can be detected

5:11 by all the other stations the collided signal may be retransmitted the last requirement is that stations can have

5:20 either carrier sensing or not carrier sensing is a specific electrical signal on the network sensing of carrier gives

5:29 the status of the channel That is busy or free next we are going to discuss multiple access

5:38 protocols one feature of Lan is that its backbone is a shared Channel or transmission link which provides all

5:46 users to access the transmission facilities it may happen that two or more stations transmitting data at the

5:54 same time causing their signals to interfere and data gets gobbled so in order to resolve these type of conflicts

6:02 a number of different medium access protocols have been developed in order to handle the bursty nature of land

6:10 burstiness means the data is transmitted irregularly in bursts rather than as continuous streams traffic asynchronous

6:19 stedium is used the asynchronous stedum mechanism is further divided into two methods

6:26 contention method or Random Access method method and deterministic Method or they are also called as controlled

6:34 methods Random Access techniques are Aloha carrier sense multiple xes or

6:41 csma then csma with Collision detection which is also called csms CD and the last one is register insertion it is not

6:50 very popular and is obsolete controlled access to landan can be performed in two types centralized or

6:59 distribute it in centralized technique Master node decides which node is to access the channel at any point of time

7:07 for example polling in distributor technique each station is given an opportunity to transmit on the channel

7:14 for example token passing method and slotted ring method this figure illustrates the

7:22 typical multiple Access Communications where a number of user stations share a transmission medium

7:30 this sharing techniques are used both in wired and wireless communication networks in wired communication

7:38 multi-drop cables are used in data networks to connect number of stations to a host computer the host computer

7:46 broadcasts information to the users on the outbound line the stations transmit information to the host using the

7:55 inbound line this system is shown in this figure a medium access control or Mac protocol

8:04 is developed for the system here the host computer issues polling messages to each stations providing it with

8:13 permission to transmit on the inbound line in radio communication several stations share two frequency bands one

8:22 for transmitting and one for receiving in Satellite Communications each station is assigned a channel in an Uplink

8:30 frequency band that it uses to transmit to the satellite the satellite sends

8:37 back the signals on different frequency band called down link frequency

8:43 band first technique is Aloha the Aloha protocol is a contention

8:51 protocol which was developed at the University of Hawaii in early 1970s and was originally developed for packet

9:00 Radio Networks however it is applicable to any shared transmission medium in a system

9:08 when many users try to send messages to each other using a common broadcast channel Random Access or contention

9:16 techniques are used Random Access means there is no exact time for any station to transmit data this scheme is very

9:24 simple and is asynchronous it is asynchronous because there is no proper coordination among the transmitting

9:32 stations the basic idea of Aloha system is useful to any system in which coordinated stations are competing for

9:41 the use of a single shared Channel when a station sends data another station may

9:48 try to do so at the same time the data from the two stations May Collide and get scbl if there is a collision then

9:56 each station would simply wait for a random period of time before try again the Aloha system is of two types Pure

10:05 Aloha which does not require Global time synchronization and slotted loha which

10:12 requires time synchronization will first discuss Pure Aloha the original Aloha protocol is

10:21 called Pure Aloha in this the whole idea is that each station can send a frame

10:28 whenever it has a frame to send as there is only one channel available to share

there is the possibility of collision

- 10:36 between frames from different stations the Pure Aloha protocol depends upon acknowledgement from the receiver when a
- 10:44 user sends a frame it expects the receiver to send an acknowledgement if the acknowledgement does not reach the
- 10:52 transmitter after the timeout period the station assumes that the frame has been destroyed
- 11:00 and resends the frame the figure shows the frame collisions in Pure Aloha as you can see when two frames are
- 11:09 transmitted at the same time there will be a collision and both will be gobbled if the first bit of a new frame overlaps
- 11:18 with just the last bit of a frame almost finished then both the frames will be completely destroyed and both will have
- 11:26 to be retransmitted again after the time of period if all the stations try to resend their frames after the timeout the
- 11:35 frames will collide again Pure Aloha dictates that when the timeout period passes each user waits a
- 11:45 random amount of time before resending its frame the randomness will help to avoid more collisions this time is
- 11:53 called as back of time TB the timeout period is equal to the maximum possible roundtrip propagation
- 12:02 delay which is twice the amount of the time required to send a frame between the two most widely separated stations 2 into TP let all the packets have the same
- 12:08 length and each station requires one time unit for transmission which is TP
- 12:17 consider any user to send packet a at the time T not if any other user B has
- 12:24 generated a packet between time T not and T not plus TP the end of a packet B
- 12:31 will collide with the beginning of packet a since in P Aloha packet a station does not listen to the channel
- 12:40 before transmitting it has no way of knowing that above frame was already
- 12:48 underway this figure shows what vulnerable periods during which packets can
- 12:56 Collide similarly if another user wants to transmit between t plus TP and t + 2
- 13:03 TP that is Packet C the beginning of packet c will collide with the end of
- 13:12 packet a thus if two packets overlap by even the smallest amount of the
- 13:19 vulnerable period both packets will be corrupted and need to be retransmitted let's
- 13:27 calculate the throughput of Pure
- 13:36 Aloha channel the throughput S of Pure Aloha is defined as an average successful traffic
- 13:45 transmitted between the stations per unit time the unit of time is slot time

13:52 which is the time required to transmit a frame assuming all frames or packets are
13:59 of same size since only one packet per slot can be transmitted the maximum value
of
14:07 is 1 when collisions happen some of the packets are lost and part of available
Channel time is wasted the resulting
14:16 value of s is less than one to find the throughput of transmission Channel
14:23 suppose the probability P_K that K packets are generated during a given slot time
follows a poisson distribution
14:32 with the mean G per packet time is given by this equation the throughput S is then
just
14:40 the offered load G times the probability of a transmission being successful s is
14:47 equal to $G P$ where p is equal to probability that a packet does not suffer a
14:55 collision the probability of no other traffic being initiated during the entire
vulnerable period is thus given
15:04 by P is equal to e^{-2G} therefore throughput s is equal to G
15:12 multiplies e^{-2G} the maximum throughput occurs at when G is equal
15:22 to 0.5 therefore for Pure Aloha the best Channel utilization that can be achieved is
around
15:31 18% no synchronization is required for Aloha each station can transmit a packet
15:38 whenever it has one the disadvantage of this scheme is that it is inefficient because
the maximum Channel utilization
15:46 is only 18% of the total available capacity next is slotted Aloha in
15:55 slotted Aloha the total Channel time is divided into discrete time slots and the
stations can transmit only at
16:03 specific instance of time these time slots are exactly equal to the packet
transmission time all stations are then
16:11 synchronized to these time slots so that whenever a station has a packet it must
synchronize exactly with the next
16:20 possible time slot the benefit of this scheme is that the wasted time due to
collisions can be reduced to one packet
16:28 time or we can say that vulnerable period is reduced to half transmission attempts
for four Network users and
16:36 random retransmission delays for colliding packets in slotted aloha are shown
in this
16:44 figure there are few assumptions in case of slotted aloha all the frames are of
the same size
16:52 time is divided into equal size slots a slot equals the time to transmit one
16:59 frame nodes start to transmit frames only at the beginning of a slot nodes are
synchronized if two or more nodes

17:08 transmit in a slot all nodes detect Collision before the slot ends let's discuss the throughput of slotted Aloha

17:17 channel in slotted Aloha the packets arrive in a synchronized fashion the probability of single transmission

17:25 during a slot time is p not is equal to e^{-G} therefore

17:31 throughput s is equal to G multiplied with e^{-G} the maximum throughput occurs at G equal to 1 which

17:40 is twice that of Pure Aloha this means that the best Channel utilization that can be achieved is around

17:50 36.8% next technique is Carrier sense multiple AIS that is csma Carrier sense multiple AIS is a

17:59 probabilistic media Access Control protocol in which a station confirms the absence of other traffic before

18:08 transmitting a data on a shared transmission medium carrier sense means that a transmitting station tries to find out whether another transmission is

18:17 in progress before starting a transmission that is it tries to identify the presence of a carrier

18:24 signal from another station before starting to transmit if if a station senses a carrier the station waits for

18:32 the transmission in progress to finish before starting its own Transmission in other words csma scheme is based on the

18:40 principle of sense before transmit or listen before talk multiple access means that more than one stations may send and

18:49 receive on the shared transmission medium the vulnerable time for csma is the propagation time TP the propagation

18:59 time is the time required for a packet to propagate from one end to the other end of the medium now look at this

19:07 figure in which station A transmits a frame at time T_1 which reaches the rightmost station D at time $T_1 + TP$ this

19:17 figure shows the vulnerable time for csma when a station senses the medium to

19:24 be idle it transmits the packet by using one of these approaches nonpersistent csma one persistent csma

19:33 and P persistent csma nonpersistent csma is a non-aggressive transmission protocol

19:40 when the transmitting station has a data to transmit it first senses the medium if the medium is Idle then it transmits

19:48 the data immediately but if the medium is busy then it waits for a random period of time during which it does not

19:55 sense the transmission medium before repeating the whole process that is sensing the transmission medium for idle

20:03 or busy again this scheme reduces the collisions and which results in higher medium

20:11 throughput next is one persistent csma one persistent csma is an aggressive

20:18 transmission algorithm when the transmitting station is ready to transmit data it senses the medium for

20:24 idle or busy if the medium is Idle then it transmits the data immediately but if the medium is busy then the station

20:32 senses the medium continuously until it becomes idle then it transmits the message unconditionally that is with

20:40 probability equals to 1 in case of any Collision the sender waits for a random period of time and attempts to transmit

20:48 again unconditionally that is with probability equals to 1 one persistent csme is used in CSM CD systems including

20:58 it ethernet next is p persistent CSM this is an approach

21:05 between non-persistent and one persistent csma access schemes when the transmitting station has a data to

21:12 transmit it then senses the transmission medium for idle or busy if the medium is Idle then it transmits the data

21:20 immediately but if the medium is busy then it senses the medium continuously until it becomes idle then transmits a

21:28 data frame with probability P if the station does not transmit data frame that is the probability of this event is

21:37 1 minus P then it waits until the next available time slot P persistent csma is

21:44 used in CS smca systems including Wi-Fi and other packet Radio Systems next technique is Carrier sense

21:53 multiple aess with Collision detection that is CSM CD carrier sense multiple access with

22:00 Collision detection is a media access control method used mostly in lens which are using early ethernet technology it

22:10 uses a carrier sensing technology in which a transmitting station detects other signals in the medium while

22:18 transmitting a frame and stops transmitting that frame transmits a jamming signal in case of collisions and

22:26 then waits for a random period of time time before trying to resend the frame csms CD is a modification of pure

22:35 carrier sense multiple access that is CSM scheme CSM CD is used to improve csma performance by terminating

22:44 transmission as soon as Collision is detected the base protocol is that a station with a message to send must

22:52 monitor the channel to see if any other station is sending if another station is sending the second station must wait or

23:01 23 minutes, 1 second defer until the sending station has finished then it may send its message if no station was sending at the time that

23:10 it first listened the station may send its message immediately the term carrier sensing

23:17 indicates this listening before transmitting Behavior if two or more stations have messages to send at the

23:25 same time and they are separated by a significant distance is on the bus each May begin transmitting at roughly the

23:33 same time without being aware of the other station the signals from each station will superimpose on the channel

23:42 and is gobbled beyond the decoding ability of the receiving station this is termed as Collision let's discuss

23:51 Collision detection procedure in csms the following procedure is used to resolve a detected Collision

23:59 the procedure is complete when retransmission is initiated or the retransmission is aborted due to

24:07 numerous collisions continue transmission with a jam signal instead

24:13 of frame header data or CRC until minimum packet time is reached to ensure

24:20 that all receivers detect the Collision increment re transmission counter was the maximum number of

24:29 transmission attempts reached if so about transmission calculate and wait random

24:36 back of period based on number of collisions re-enter main procedure at stage

24:45 one because of quick termination of transmission time and bandwidth is saved therefore CSM CD is more efficient

24:53 than Aloha slotted Aloha and csma CSM CD Network Works work best on bus

25:01 25 minutes, 1 second multi-point topology with busy asynchronous transmission all stations are attached to One path and monitor the

25:09 signal on the channel through transceiver attached to the cable csmd has totally decentralized control and is

25:16 based on contention access csms CD supports both baseband and broadband system csmd offers four options in terms

25:26 of bit rate signaling method and maximum electrical cable segment length these

25:32 are 10 Bas 5 10 Bas 2 10 broad 36 and 1 base 5 the numeric field in the

25:42 beginning indicates the bit rate in MVPs the middle term indicates the type

25:49 of signaling system that is baseband or Broadband the numeric field in the end indicates the electrical cable segment

25:56 length in multi of 100 m Manchester signal code is used at the basement

- 26:03 level of transmission in Broadband transmission different phase shift keying is used to convert the Manchester
- 26:11 encoded signal into analog form so friends this was all about Mac
- 26:17 Subler and Mac protocols hope all the concepts are clear to you hope to see you in the next lecture till then
- 26:26 goodbye thank you